

Blue = research Yellow = Setup/creating idea, Green = debugging, Red = developing/coding

TECHNOLOGY STUDENT ASSOCIATION PLAN OF WORK

Date	Task	Time involved	Team member responsible (student initials)	Comments
12/14/25 - 12/20/25 1.	Develop/Formulated a solution to remove barriers to people with vision or hearing disabilities for the 2026 competition.	5 hours	S.G, A.J.J, A.T, S.S, S.S.S, A.P	Meet as a group digitally on a call to brainstorm and research possible solutions to increase accessibility to people with hearing and sight disabilities. Viewed successful projects like google lookout for inspiration. Put potential ideas on a shared document.
12/25/25 - 12/26/25 2.	Decide on what solution we as a group will work in order to remove barriers to people with hearing or vision disabilities for the 2026 TSA software development compition.	2 hours	S.G, A.J.J, A.T, S.S, S.S.S, A.P	Meet as a group digitally on a call to narrow down our solution for the 2026 TSA Software development prompt settled on a real time ASL alphabet translator which converts hand signs into text and speech.
1/02/26 3.	Research what specific ASL database we would use for our project.	3 hours	S.G, A.P, S.S, A.J.J	Researched various ASL alphabet data sets on sites like Kaggle and GitHub. Found datasets on Kaggle with over 87 thousand images, as well as "MS-ASL" data set on Microsoft.
1/03/26 - 1/09/26 4.	Research what Python Libraries are needed in order for our solution	8.25 hours	S.G, A.J.J, A.T, S.S, S.S.S, A.P	Used various credible sources to find libraries for our solution 1. How will our solution obtain camera access? 2. What Libraries allow text-to-speech? 3. What Libraries allow movement recognition? 4. How will our data be analyzed
1/24/26-1/26/26 5.	Research what Neural Networks would be the best for our solution	4 hours	A.P, S.G, S.S.S, A.T, A.J.J	Researched various Nuerl Network models for ASL recognition and adaptability. Compared various Nuerl Networks for optimized speed and ASL accuracy, ultimately settled for googles EfficientNet-B3 from Google
2/6/26 6.	Downloaded necessary files, and Pyton Libraries needed for our project	2 hours	S.G, S.S.S, S.S, A.T, A.J.J	Downloaded neccessary components, and libraries of our project like pytsx3, pillow, opencv-python, and timm. As well as set up visual studio code and downloaded collaboration software which allows multiple group members to colaborate on code.

Advisor signature

Meghan W

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2/7/26 1.	Research into physical detection of signs	3 hours	S.G, A.P, S.S, A.T, S.S.S	Researched documentation for live tracking as well as explored the capabilities for MediaPipe for robust hand tracking.
2/8/26 - 2/14/26 2.	Utilizing complex libraries to train said program to be able identify different signs presented in front of a camera.	10 hours	S.G, A.P, S.S, A.T, S.S.S	Developed ASL alphabet detection model with MediaPipe Hands library to make the hand tracking system in which it can be translated into text and later into speech.
2/15/26 - 2/21/26 3.	Text to speech integration	7 hours	S.G, A.P, S.S, A.T, S.S.S	The text-to-speech integration uses the recognized words from the ASL translator and converts them into spoken words using a background speech engine. This background speech engine works on a separate thread so that the sound produced does not interfere or slow down the processing of the webcam.
2/28/26 - 3/2/26 4.	Video pipline development	6 hours	S.G, A.P, S.S, A.T, S.S.S	Incorporated MediaPipe with OpenCV webcam to create a real time hand detection system, so ASL alphabets can be easy detected and translated from text to speech.
3/10/26 - 3/12/26 5.	Debugging core functionalities	6 hours	S.G, A.P, S.S, A.T, S.S.S	Problems encountered in initialized the webcam were fixed later during early runtime. Achieved by checking cap.isOpened().
3/24/26 - 3/25/26 6.	Creation of an easy use User Interface	3 hours 30 minutes	S.G, A.P, S.S, A.T, S.S.S	The user interface in ui.py displays the webcam feed and model output in a structured, interactive layout with sections for status, words/sentences, letter guide, and practice tools, along with buttons for actions like start, pause, and clearing data. It updates dynamically in real time based on the model's predictions.

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3/26/26 1.	Debugging the User Interface to allow for easy usage.	2 hours	S.G, A.P, S.S, A.J.J, S.S.S, A.T	During the debugging stage of the user interface the time between piplines and on screen displays were shortened drastically. Additionally corrections relating to overlapping buttons were fixed.
3/28/26 - 3/29/26 2.	Documentation of feats (requirements to run code)	4 hours	S.G, A.P, S.S, A.J.J, S.S.S, A.T	Worked on README file with directions on how to use the softwear. Additionally a fully filled color coded work log was created.
4/1/26 3.	Submission	1 hour	S.G, A.P, S.S, A.J.J, S.S.S, A.T	Documented several requirements needed for the software to run proper. Included the main language of python as well as several dependencies like OpenCy, MediaPipe, and pyttsx3. Additonally the main code can be ran through app.py file
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Meghan Wolf